

Boundary integral method for Poisson's equation with orbital data

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Currently, we have developed a boundary integral method to solve the Poisson's equation with dielectric interfaces using the orbital data as the source term. In this note, I will briefly describe the solver, and show some preliminary results.

First, as been discussed in the previous notes, the equation to solve is given by

$$\left(\frac{1}{2} \frac{\epsilon_i + \epsilon_j}{\epsilon_i - \epsilon_j} + D^T\right) \sigma(x) = -\partial_{\mathbf{n}} u_{\text{inc}}(x), \quad x \in \partial\Omega_i \cap \partial\Omega_j, \quad (1)$$

where $\sigma(x)$ is the unknown surface charge density, D^T is the transpose of the double layer potential operator, and $u_{\text{inc}}(x)$ is the incident potential generated by the orbital data, given by

$$\partial_{\mathbf{n}} u_{\text{inc}}(x) = \int_{\Omega_o} \frac{\partial G(x, y)}{\partial \mathbf{n}_x} \rho(y) dy, \quad (2)$$

where $G(x, y)$ is the Green's function of the Poisson's equation, and $\rho(y)$ is the charge density computed from the orbital data. Eq. (1) is a Fredholm integral equation of the second kind¹, by solving which we can obtain the surface charge density $\sigma(x)$, and then compute the scattered potential and field at any point in the domain. Our algorithm consists of the following steps to solve the above equation:

1. Load the orbital data and compute the charge density $\rho(y)$ in the orbital region Ω_o
2. Discretize the boundary into a set of panels, and evaluate $\partial_{\mathbf{n}} u_{\text{inc}}(x)$ on each panel
3. Construct the discretized representation of the operator D^T using the discretized boundary
4. Solve the resulting linear system to obtain the surface charge density $\sigma(x)$

We first load the wannier function stored in the `.xsf` files. For the ease of calculation, we shift the wannier function to be centered at the atom, and truncate the function in z as shown below:

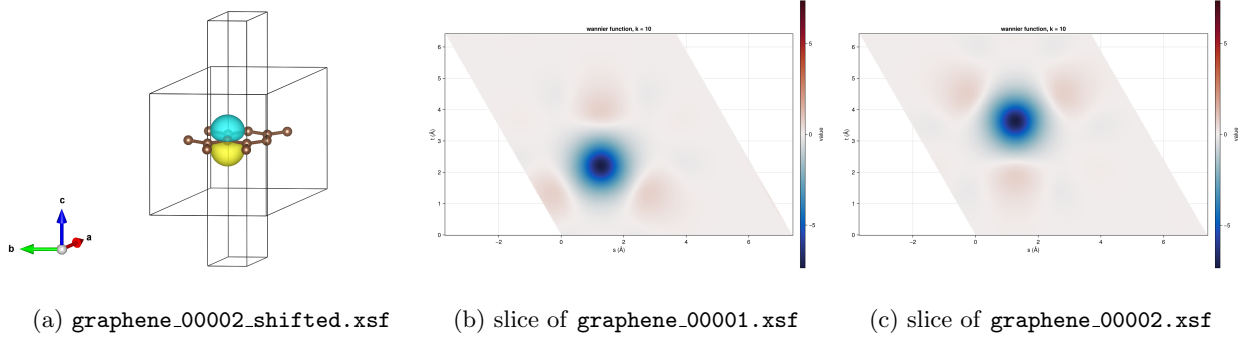


FIG. 1: Wannier functions from the `.xsf` files.

Then by multiplying the wannier functions with each other, we can obtain the charge density $\rho(y)$. In the following examples, we simply consider the density generated by the wannier function in `graphene_00002.xsf` being multiplied with itself, which is inserted into a box of size $(90, 90, 9)$ to represent the 2D material, with relative permittivity $\epsilon_r = 6$. Based on the charge density, we discretize the surface into panels, and on each panel we use a $p \times p$ Gauss-Legendre quadrature to sample the right hand side of Eq. (1). The orbital density and the panels are shown in Fig. 2, where the panels near the orbital density and edges are refined to capture the singularity of the kernel and the corner singularity, respectively.

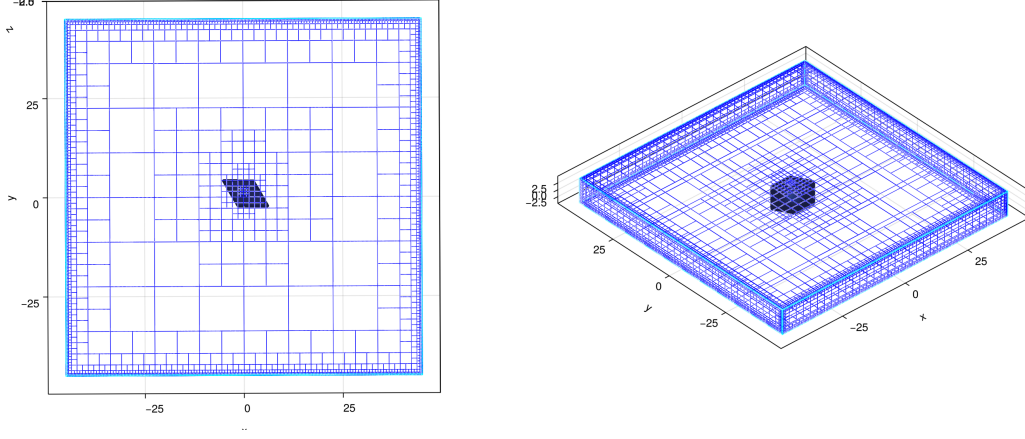


FIG. 2: Orbital density and panels on the boundary of the dielectric interface.

With the discretized surface, we constructed a discretized representation of the operator/values on the left/right hand side of Eq. (1), and solve the resulting linear system to obtain the surface charge density $\sigma(x)$, as shown in Fig. 3.

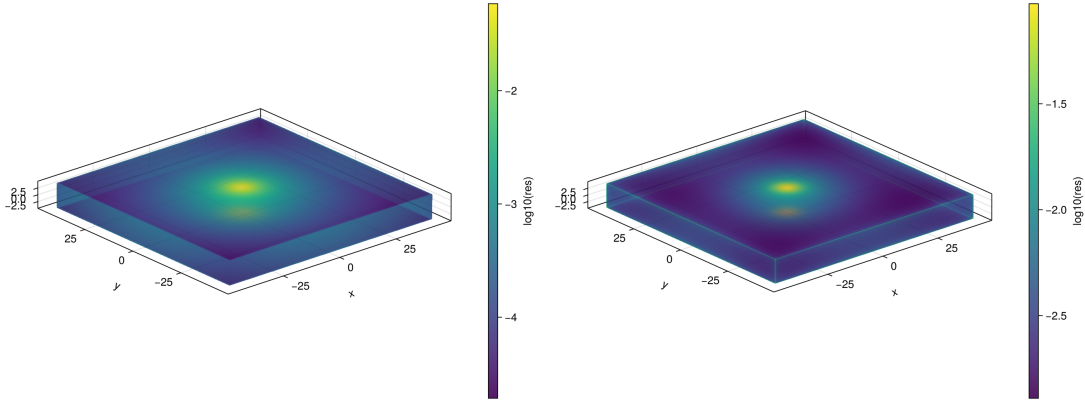


FIG. 3: RHS (left) and solution (right) of Eq. (1).

Finally, with the discretized representation of the surface charge density $\sigma(x)$, we can evaluate the potential at any point in the domain. To validate the solver, we do a self convergence test by evaluating the potential at a set of fixed points with different order of quadrature and edge refinements, as shown in Fig. 4, which shows that the potential converges exponentially with the level of the edge refinement.

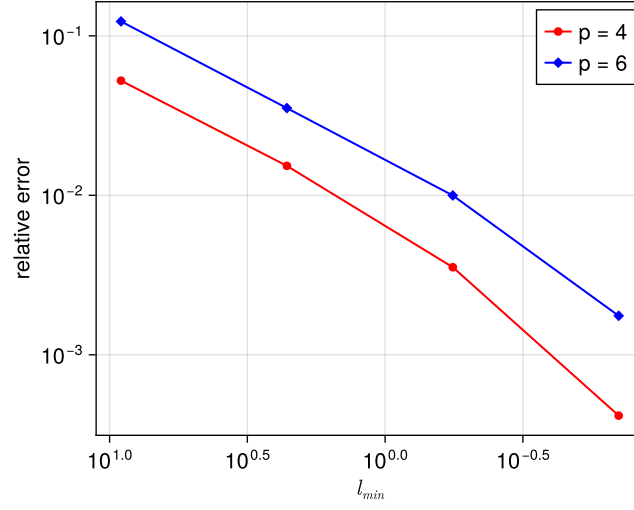


FIG. 4: Convergence of the potential at fixed points with different order of quadrature p and edge refinements l_{min} .

¹ L. Greengard and M. Moura, Acta numerica **3**, 379 (1994).